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Binding Theory

Daniel Büring

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Binding Theory

Binding Theory seeks to explain how different kinds of nominal expressions such as names, noun phrases, and pronouns have anaphoric relations among one another, and how they come to have reference to things in the world. This textbook provides a thorough and comprehensive introduction to modern Binding Theory. Starting at a very basic level, it introduces the reader to a huge variety of nominal and especially pronominal expressions from the world's languages, the ways they can be used, and current theorizing about their grammatical properties and their interpretation. Daniel Büring discusses a wide range of cross-linguistic data and theoretical approaches, and, unlike in existing introductions, pairs the discussion of syntactic facts with a detailed introduction to the semantic interpretation of binding structures. Written in a clear and accessible style, and with numerous exercises and examples, this textbook will be invaluable to graduate and advanced undergraduate students of syntax and semantics.

DANIEL BÜRING teaches linguistics at the University of California, Los Angeles. He has published various influential articles in formal semantics, syntax, and pragmatics, in particular on intonational meaning, focus, and binding theory. He has previously published *The Meaning of Topic of Focus: The 59th Bridge Street Accent* (1997).

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DANIEL BÜRING

University of California, Los Angeles

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Contents

Preface *page* ix

1 The ABC of Binding Theory 1
- 1.1 Preliminaries 1
- 1.2 Binding 4
- 1.3 Command and precedence Ⓔ 12
- 1.4 Reflexive verbs and reflexive phraseologisms Ⓔ 21

2 Interpreting indexed structures 25
- 2.1 Basics of interpretation 25
- 2.2 Enter indexing 26
- 2.3 Compositional interpretation 31
- 2.4 Extensions and alternatives Ⓔ 40

3 Domains and orientation 46
- 3.1 Binding domains in English: governing category 46
- 3.2 Orientation 58
- 3.3 Binding domains cross-linguistically 65
- 3.4 Long-distance reflexives 72
- 3.5 Some pronominal systems 75

4 Binding versus coreference 81
- 4.1 Quantified NPs and variable binding 83
- 4.2 The syntax of semantic binding 89
- 4.3 *Wh*-expressions 93
- 4.4 Summary 95
- 4.5 Semantic details Ⓔ 96

5 Other cases of semantic binding 104
- 5.1 Overture 104
- 5.2 Focus constructions 105
- 5.3 Double indexing 109
- 5.4 A new system 110
- 5.5 Verb phrase ellipsis Ⓔ 114

6 The Coreference rule 118
- 6.1 The proposal 118
- 6.2 Theoretical consequences 119

6.3 Binding Theory obviations 126
6.4 Summary: the final system 128
6.5 Extensions 130

7 Descriptive pronouns and individual concepts 143
7.1 Anaphoric pronouns that don't corefer 143
7.2 Unknown and mistaken identity 151
7.3 Descriptive NPs and indexing℮ 156
7.4 Summary 159
7.5 An extension: unexpected sloppy identity 160

8 Semantic binding and c-command 163
8.1 The weak crossover phenomenon 163
8.2 Blocking weak crossover 166
8.3 A challenge: indirect binding 174

9 Plurals 188
9.1 The semantics of plural NPs 188
9.2 Anaphoric relations between plural NPs 191
9.3 Set indexation 193
9.4 More on overlapping reference℮ 199

10 Reciprocals 203
10.1 Plural preliminaries 204
10.2 Strong reciprocity 206
10.3 The syntax of reciprocal binding 209
10.4 Alternative meanings for reciprocal sentences 213
10.5 Reflexives and reciprocals 220

11 Exempt anaphora and reflexivity 222
11.1 Introducing exempt anaphora 222
11.2 Conditions on exempt anaphora 225
11.3 On the notion of higher coargument 227
11.4 Reflexivity Theory 235
11.5 Towards a cross-linguistic perspective 242

12 Binding and movement 244
12.1 Argument movement 244
12.2 *Wh*-movement 246
12.3 Analytical options 248
12.4 An apparent case of binding after *wh*-movement 254
12.5 A real case of interaction of Ā-movement and BT? 256
12.6 Binding without binders 260

Bibliography 265

Index 277

Preface

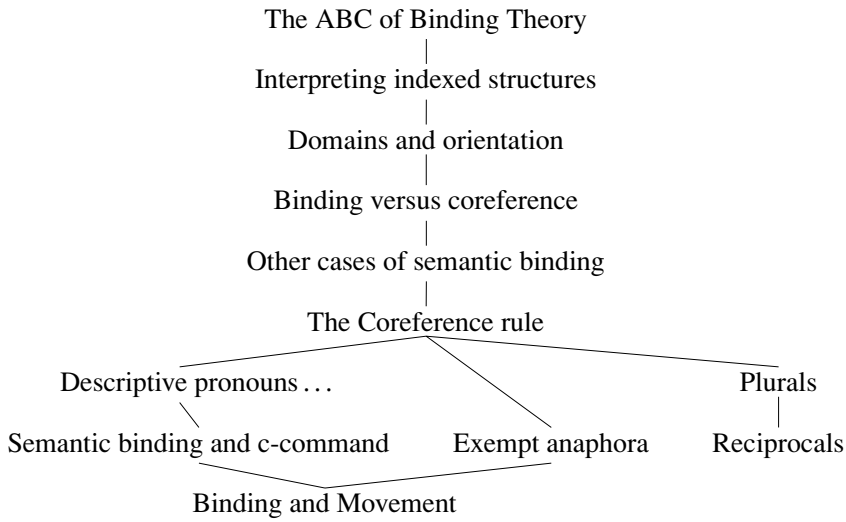
This book presents a comprehensive treatment of the syntax and semantics of binding. It is meant to fill the gap between existing introductory texts, both semantic and syntactic, and the rich primary research literature on the topic. If you work your way through this book, you should be able to read and understand almost any of the works mentioned in the references.

There are at least two reasons why I thought such a book may be useful. First, Binding Theory figures prominently in a vast amount of works, either as the main research topic, or, perhaps even more frequently, as a diagnostic for constituency, derivational history, and other abstract aspects of grammatical analysis. I felt that an accessible survey of some of the more recent insights into the nature of binding would benefit both those who read those studies, as well as those who want to undertake them in the future.

Second, by its very nature, Binding Theory involves an equal amount of syntax and semantics. As such, it recommends itself as the topic for an advanced level textbook. There is, I believe, no insightful syntactic analysis without a solid semantics to access its adequacy; in any event, there certainly can't be any insightful analysis of the syntax of binding without a semantics to accompany it. The present book, therefore, is an introduction to doing syntactic and semantic analysis side by side. It attempts to show you how to do semantically realistic (or responsible) analysis; it will also show you how, at least in some cases, figuring in the semantics carefully may solve some problems that would seem recalcitrant from a purely syntactic point of view. It's good old *divide et impera*.

The book is organized as follows: the first six chapters develop, in incremental steps, the basic system of NP classification, indexing, and interpretation. They each crucially build and expand on the content of the preceding ones, and should be tackled in that order. Chapters seven through twelve then extend the basic system in various, sometimes opposite, directions, and can be accessed mostly independently of each other; this structure is schematized in the chart below.

Within chapters, certain sections are marked as Ⓔ, for "extension"; these often contain more advanced and demanding material, and can be skipped without loss of coherence for later chapters (except possibly the Ⓔ-parts therein).



I have attempted to introduce explicitly every piece of machinery used in the analysis, and make all assumptions explicit. I have also included a fair number of exercises, especially in the earlier chapters, that should help to master the material, but also to discover problems and open ends. Despite that, I think that a certain familiarity with linguistic argumentation, as well as with formal syntactic and semantic analysis is required to read this book. Most introductory textbooks should provide the necessary background.

When Cambridge University Press invited me to write this book, I had taught 'The Syntax and Semantics of Binding Theory' at a couple of summer schools, and the plan was essentially to flesh out the existing course materials. In the process of writing the book, more and more literature made its way into these materials, and the scope of the book extended considerably. Still, this book is not a natural history of binding phenomena, especially not cross-linguistically, and makes no claim to do justice to the vast theoretical and especially descriptive literature, of which only a fraction is taken into consideration here. While I tried to use examples from many different languages, where I had sufficient sources, the primary language analyzed is English. And even there, I found that the reported judgments are often very subtle and highly controversial. I sincerely believe now that much more systematic primary work on establishing a firm data base needs to be done; as it is, I mostly report the data as given in the literature, pointing out points of controversy, and occasionally supplementing native speaker judgments I elicited.

There are also some areas that are omitted altogether in this book, mostly for reasons of space, among them the diachronic changes in anaphoric systems (van Gelderen [2000]; Keenan [2002]), as well as their acquisition in young children (Wexler and Manzini [1987], a.o.). Furthermore, older theoretical approaches to Binding Theory are not discussed, though they might often

facilitate understanding more recent approaches (I recommend the first chapters of Kuno [1987] for an excellent overview).

More people than I can mention here have helped me in the process of writing this book. I'd like to thank in particular Ed Keenan and Philippe Schlenker, my colleagues here at ULCA, for their input, and Daniel Hole and Chris Potts for their extremely detailed comments and suggestions; thanks also to Christina Kim for helping with the final proofs. Special thanks go to Summer Kern, my *Herzallerliebste*, for her support, encouragement, and patience, and for always (perhaps reluctantly) being willing to double-check yet another sentence or two.

1 The ABC of Binding Theory

1.1 Preliminaries

1.1.1 Reference, coreference, and indexing

What is Binding Theory (BT) about? To a first approximation, BT restricts the distribution of NPs (or DPs, if you prefer) that have the same *referent* (starting with chapter 4, we will add non-referential NPs to the picture, which will be ignored until then). We will indicate sameness of reference, *coreference* for short, by *coindexing*; that is, coreferent NPs carry the same *index*, for which we use integers throughout. Thus in (1.1), the NP *the baroness* and the NP *she* are coindexed, which signals that they are coreferent, which in turn means that they have the same referent – they refer to the same person or thing – namely the actual baroness in flesh and blood:

(1.1) After $[_{NP}$ the baroness$]_1$ had visited the lord, $[_{NP}$ she$]_1$ left the house.

Note that on this understanding, BT is relevant for nominal categories only, and only for the maximal projections, i.e. NPs.[1] As a convention we assume that two NPs corefer if *and only if* (iff) they are coindexed. Contra-indexing (or lack of an index on either NP) indicates non-coreference. This is illustrated in (1.2):

(1.2) (a) After $[_{NP}$ the baroness$]_2$ had visited the lord, she$_2$ left the house. (she=the baroness)
(b) After $[_{NP}$ the baroness$]_1$ had visited the lord, $[_{NP}$ she$]_2$ left the house. (she≠the baroness)

It should be noted that the actual choice of integer is irrelevant; (1.1) expresses the same *coreference pattern* as (1.2a) (as would any sentence in which both occurrences of the index are replaced by the *same* integer). An NP marked 1 is in no sense prior, higher, or superior to one marked 2. All that matters is which NPs have the same index, and which do not.

[1] The latter aspect I consider a genuine fact about Binding Theory. On the view pursued here, indexing on non-maximal projections (e.g. signalling specifier-head agreement or head-movement dependencies) simply is not subject to Binding Theory and should be kept separate from it. As for the former aspects, though there are sentential and adverbial (i.e. PP-) anaphors, little work on their distribution has been done, and we will ignore them here (see e.g. Hegarty *et al*. [2001] and the references therein).

In traditional grammars, the NP *the baroness* in (1.1) is referred to as the *antecedent* of the pronoun *she*. We adopt the following:

(1.3) Definition: Antecedent
A is the *antecedent* of B iff (if and only if) (i) A precedes B, and (ii) A and B corefer.

By our convention, an NP will be coindexed with its antecedent (if it has one). This holds for coreferring NPs within a single sentence, and across sentences. The latter, however, are usually not subject to Binding Conditions of the sort discussed here.[2]

1.1.2 The basic data

Restricting our attention to singular NPs for the time being, two NPs in a given sentence will show one of three logically possible coreference relations (Reinhart, 1983a: 29):

(1.4)
(a) *obligatory coreference:* Zelda bores herself.
(b) *obligatory non-coreference:* She adores Zelda's teachers.
(c) *optional coreference:* Zelda adores her teachers.

Given what was said before, grammatical representations for these will look like in (1.5):

(1.5)
(a) Zelda_1 bores herself_1.
(b) She_8 adores Zelda_{15}'s teachers.
(c) Zelda_4 adores her_4 teachers. *or*
Zelda_4 adores her_7 teachers.

Ungrammatical representations for (1.4a) and (1.4b) are given in (1.6):

(1.6)
(a) $*\text{Zelda}_1$ bores herself_2.
(b) $*\text{She}_8$ adores Zelda_8's teachers.

It will be convenient to summarize patterns as in (1.5) and (1.6) as shown in (1.7), whose logic should be transparent:

(1.7)
(a) Zelda_1 bores $\text{herself}_{1/*2}$.
(b) She_8 adores $\text{Zelda}_{15/*8}$'s teachers.
(c) Zelda_4 adores $\text{her}_{4/7}$ teachers.

The key insight captured in BT is that the (un)availability of coreference between two NPs crucially depends on two factors:

[2] See e.g. Grosz *et al*. (1995); Gundel *et al*. (1993); Walker *et al*. (1998) and the references therein for some discussion of trans-sentential anaphora.

- the morphological shape of the NPs
- the structural relation between the NPs

This is not meant to exclude the possibility of additional factors that influence coreference options (which will be discussed especially in chapters 3 and 11). First, however, we will introduce the relevant NP-types of English and then, in turn, explore and characterize the syntactic configurations in which they require, allow, or disallow coreference.

1.1.3 Three types of NPs

Virtually all approaches to BT in English distinguish three types of NPs by (mostly) morphosyntactic criteria. These are illustrated in (1.8a–1.8c):

(1.8)
(a) reflexives and reciprocals ('anaphors'):
himself, herself, itself, themselves, myself, yourself, ourselves, yourselves
each other, one another
(b) non-reflexive pronouns ('pronominals'):
he, she, it, him, her, I, us, you, me, his, your, my, our
(c) full NPs including names ('r-expressions'):
the baroness, Peter, this, a disinherited Russian countess . . .

In parentheses I have given the terms for these categories as used in the influential work of Chomsky (e.g. 1981) and his school: anaphor, pronominal, and r-expression (with *r* reminiscent of 'referential'). For the first two, a cautionary remark is in order, because they unfortunately provide potential for confusion: traditionally the term *anaphor* (often with the plural *anaphors* rather than *anaphora*) is used for any NP, reflexive or not, that has an antecedent. Likewise, the term *pronominal* invites confusion with the traditional notion of *pronoun*, which applies to reflexive and non-reflexive pronouns alike. We will thus stick to the terms 'reflexive/reciprocal', 'non-reflexive pronoun', and 'full NP' in the remainder of this book.

We will now motivate this tripartition, starting with reflexives versus the rest (reciprocals, being necessarily plural, will not be discussed until chapter 10). Consider the sentences in (1.9):

(1.9)
(a) That it rains bothers Peter.
(b) That it rains bothers her/him.
(c) ∗That it rains bothers himself/herself.

All these sentences contain but one referential NP (the expletive *it* is of no interest to BT, since it lacks a referent – and perhaps semantic content in general). We can thus omit the indexing for expository convenience, given that no coreference is involved. We simply observe that reflexives cannot occur in this configuration, while both non-reflexive pronouns and full NPs can.

Table 1.1 *Distribution of the three NP-types*

configuration	ex.	reflexive	non-reflexive	full NP
no antecedent	(1.9)	∗	ok	ok
non-local antecedent	(1.11)	∗	ok	∗
local antecedent	(1.10)	ok	∗	∗

Inversely, only reflexives, but neither non-reflexives nor full NPs, are permitted in (1.10):

(1.10) (a) ∗Peter$_3$ watches Peter$_3$ in the mirror.
(b) ∗Peter$_3$ watches him$_3$ in the mirror.
(c) Peter$_3$ watches himself$_3$ in the mirror.

(Note that the two occurrences of *Peter* in [1.10a] are coindexed, indicating that we speak about the same Peter. The sentence is presumably acceptable if I point at a different Peter upon using the names, just as [1.10b] is of course grammatical if the pronoun is not coindexed with the name.)

Let us finally turn to the difference between non-reflexive pronouns and the rest, illustrated by way of the sentences in (1.11):

(1.11) (a) ∗Carla$_4$ thinks that I hate Carla$_4$.
(b) Carla$_4$ thinks that I hate her$_4$.
(c) ∗Carla$_4$ thinks that I hate herself$_4$.

Here, reflexives pattern with full NPs, and in contradistinction to non-reflexive pronouns. Note that the difference between (1.10) and (1.11) is not the absence versus presence of an antecedent (there is one in each), but seems to be one of syntactic *locality*: the antecedent NP is within the same clause as the anaphor in (1.10), but in a higher clause in (1.11). We summarize these (preliminary) results in table 1.1. What is clear from this table is that at least this three-way distinction needs to be recognized to distinguish correctly the coreference options of NPs in English. Notice also that reflexive and non-reflexive pronouns seem to be in complementary distribution. We will now characterize the conditions for coreference for the three types of NPs in turn.

1.2 Binding

1.2.1 Reflexive and non-reflexive pronouns

We observed above that reflexive pronouns require an antecedent, and an antecedent within their local clause at that. This is illustrated in more detail in (1.12):

(1.12) (a) ∗That it rains bothers himself/herself. (no antecedent)
(b) ∗Carla$_4$ thinks that I hate herself$_4$. (non-local antecedent)
(c) Peter$_2$ watches himself$_2$ in the mirror. (local antecedent)

Turning now to non-reflexive pronouns, recall that they can occur with or without a sentence-internal antecedent, cf. (1.13), as long as the antecedent is not in the same local clause, cf. (1.13c):

(1.13) (a) That it rains bothers him/her. (no antecedent)
(b) Carla$_4$ thinks that I hate her$_4$. (non-local antecedent)
(c) ∗Peter$_3$ watches him$_3$ in the mirror. (local antecedent)

Based on these data we formulate our first version of the *Binding Conditions*:

(1.14) Binding Conditions (preliminary)
(A) A reflexive pronoun must have an antecedent within its local clause.
(B) A non-reflexive pronoun must not have an antecedent within its local clause.

(1.15) Ancillary definition:
α is within ϕ's *local clause* if α and ϕ are dominated by the same set of clausal nodes (S, $\bar{\text{S}}$, IP, CP, TP, AgrP . . .).

Exercise 1.1

In the following sentences, Φ designates an NP with the index given. For each sentence, determine by intuition what Φ can/must be (there may be more than one option in some cases). Then give the local clause and the antecedent for Φ and demonstrate that the Binding Conditions in (1.14) are met (example: Φ_3 in [1.16a] must be *himself*, its local clause is the matrix S/IP, and its antecedent is *Peter*, which is, correctly, in the same local clause):

(1.16) (a) Peter$_3$ watches Φ_3 in the mirror.
(b) Masha$_5$ believes that the swamp elks admire Φ_5.
(c) Masha$_5$ believes that [the swamp elks]$_{16}$ admire Φ_{16}.
(d) Masha$_5$ introduced Φ_5 to the swamp elks.
(e) Hermann$_8$ tried to be nice, and Gallia quite liked Φ_8. Now Φ_8 and Gallia go out to see a mud wrestling show.
(f) Masha$_5$ mentioned a swamp elk that was important to Φ_5.
(g) Φ_1's manager takes care of Cecilia$_1$'s business.
(h) Φ_1 takes care of Cecilia$_1$'s business.

1.2.2 Binding and binder

Before going on, we need to refine our previous treatment in one small but significant way. To see why, consider (1.17):

(1.17) (a) Carlotta$_{11}$'s dog accompanies her$_{11/6}$ to kindergarten.
(b) ∗Carlotta$_{11}$'s dog accompanies herself$_{11/6}$ to kindergarten.

The judgments in (1.17) are the reverse of what the Binding Conditions lead us to expect: *Carlotta* is clearly in the same local clause as *her*/*herself*, yet we

have to choose a non-reflexive pronoun to express coreference. This is in marked contrast to our earlier example (1.10), repeated here, which led to the formulation of the Binding Conditions above:

(1.18) Peter$_3$ watches himself$_3$/*him$_3$ in the mirror.

One difference is that *Peter* and *himself* in (1.18) are *clausemates*, whereas *Carlotta* and *her(self)* in (1.17) are not – *Carlotta* is the possessor to the subject, but only the subject and *her(self)* are clausemates. We can flesh out the notion 'clausemate' in various ways, e.g. as 'be arguments to the same predicate' (here: *watch*), or 'be immediate constituents of the same clause,' with subtly different results, as we will discuss immediately in sections 1.2.4 and 1.3.

Postponing a precise definition of clausemate, let us say that only an antecedent which is a clausemate to an NP can be a *binder* for that NP:

(1.19) Binding (preliminary): NP$_1$ binds NP$_2$ if and only if (iff)
(a) NP$_1$ and NP$_2$ are coindexed
(b) NP$_1$ precedes NP$_2$
(c) NP$_1$ and NP$_2$ are clausemates.
Then NP$_1$ is the *binder* of NP$_2$, and NP$_2$ is *bound* (by NP$_1$).

(1.19a) and (1.19b) are the same as in the definition of antecedent in (1.3) above, but clause (1.19c) is added. A binder, then, is simply an antecedent that is a clausemate of the bindee. We now replace the notion of 'have an antecedent' with the notion of 'be bound' in the Binding Conditions:

(1.20) Binding Conditions (still preliminary):
(A) A reflexive pronoun must have a binder within its local clause.
(B) A non-reflexive pronoun must not have a binder within its local clause.

In (1.18), repeated in (1.21a) below, *Peter* qualifies as a binder with respect to the pronoun in the object position of *watch* – it is coindexed with it, precedes it, and, being the subject of *watch*, is a clausemate. Hence Binding Condition A licenses a reflexive in object position, and Binding Condition B prohibits a non-reflexive. All's well:

(1.21) (a) Peter$_3$ watches himself$_3$/*him$_3$ in the mirror.
(b) Carlotta$_{11}$'s dog accompanies her$_{11}$/*herself$_{11}$ to kindergarten.

In the formerly problematic example (1.17), repeated in (1.21b) above, *Carlotta* is not a binder to the pronoun in the object position of *accompany* (though it is an antecedent); it is coindexed with it, and precedes it, but, being a modifier to *dog* rather than an argument to *accompany*, it fails on the clausemate condition in the definition of binder (1.19c). Binding Condition A thus prohibits a reflexive, and Binding Condition B allows a non-reflexive.

1.2.3 Full NPs

Turning now to full NPs, we observed that they cannot occur with a sentence internal antecedent at all, regardless of whether the antecedent occurs within the same local clause or not. The relevant data are repeated here:

(1.22) (a) That it rains bothers Peter. (no antecedent)
(b) ∗Carla$_4$/she$_4$ thinks that I hate Carla$_4$. (non-local antecedent)
(c) ∗Peter$_3$/he$_4$ watches Peter$_3$ in the mirror. (local antecedent)

The question that comes up is whether full NPs are allergic to antecedents, or just binders. To decide that question we have to look again at a case in which an NP antecedes a full NP without actually binding it, for example (1.23):

(1.23) (a) Her$_{11}$ dog accompanies Carlotta$_{11}$ to kindergarten.
(b) ?Carlotta's$_{11}$ dog accompanies Carlotta$_{11}$ to kindergarten.
(c) Carlotta's$_{11}$ dog accompanies the little darling$_{11}$ to kindergarten.

The pronoun in (1.23a) antecedes the full NP with no loss in acceptability. And even another full NP can, as in (1.23b), which is slightly degraded due to the repetition of the name, but head and shoulders above (1.22b); and (1.23c), which features an *epithet*, i.e. a definite NP which is coreferential with, though different in descriptive content from, its antecedent, is impeccable.

We conclude that, just as in the principles governing the coreference options of pronouns, the principle responsible for full NPs must make reference to the notion of binding, rather than antecedence:

(1.24) Binding Condition C: A full NP must not be bound.

I should like to point out here that the judgments in (1.23), while widely accepted, are not uncontroversial. Generally, name–name cases (*Peter*$_3$ . . . *Peter*$_3$) seem more acceptable than pronoun–name cases (*he*$_3$ . . . *Peter*$_3$) and for many speakers approach the degree of acceptibility found in examples like (1.23b) (cf. e.g. Bach and Partee [1980], note 11; Evans [1980]:356 a.o.). This can be seen as a phenomenon outside of grammar (after all, in the double name cases, the coreferential reading is the only way to interpret the sentence at all, while in the pronoun–name cases, there is a host of grammatical non-coreferent readings) or as a fact about BT proper, suggesting that Binding Condition C should only ban full NPs from being bound by a *pronoun* Bach and Partee [1980]; Keenan [1974]; for further discussion see also Bresnan [2000], Lasnik [1986], as well as chapter 6. We will, for the time being, assume these cases to be unequivocally bad.

1.2.4 C-command

Before closing, we need to generalize the notion of binding slightly. As it stands, Binding Condition C does not exclude (1.22b), repeated here:

(1.25) ∗Carla$_4$/she$_4$ thinks that I hate Carla$_4$.

The reason is that (the first occurrence of) *Carla*/*she* in (1.25) doesn't *bind* the second in the technical sense defined in (1.19), because they are not clausemates:

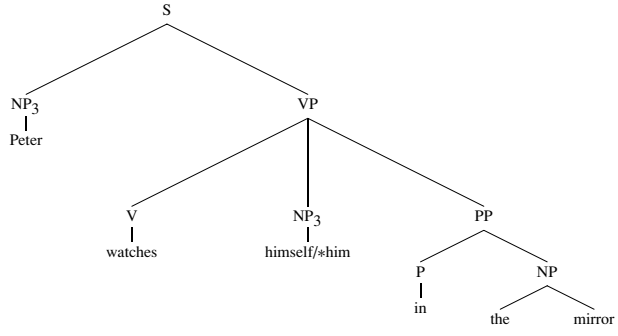
they are not immediate constituents of the same clause, nor are they arguments to the same verb (*think* versus *hate*). We therefore replace the notion of clausemate by a more general, asymmetric, notion, that of *c(onstituent)-command* [Reinhart, 1976]:

(1.26) Node A *c-commands* node B in a phrase marker iff
(a) neither dominates the other, and
(b) every (branching) node that dominates A also dominates B[3]

(1.27) Binding (revised, still preliminary): NP_1 binds NP_2 iff
(a) NP_1 and NP_2 are coindexed
(b) NP_1 precedes NP_2
(c) NP_1 c-commands NP_2
Then NP_1 is the *binder* of NP_2, and NP_2 is *bound* (by NP_1).

Let us first verify how these new definitions subsume the old ones. Take (1.21a), repeated here; a phrase structure tree for this sentence will have the essential constituency shown in (1.28):

(1.28) $Peter_3$ watches $himself_3/{*}him_3$ in the mirror.

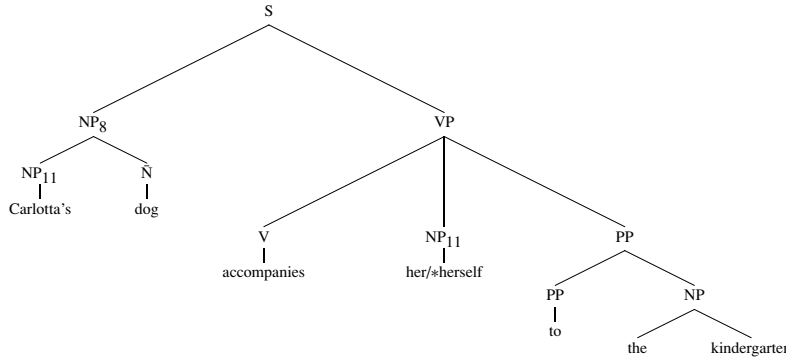


The only (branching) node dominating $[_{NP}Peter]_3$ is S, which means that $[_{NP}\,Peter]_3$ c-commands VP and everything dominated by VP, including $[_{NP}$ *himself/him*$]_3$. Thus $[_{NP}Peter]_3$ is a binder for $[_{NP}himself/him]_3$, and, given that it is in the same local clause, it is correctly predicted that the latter has to be a reflexive, rather than a full NP or a non-reflexive pronoun.

Contrast this with (1.21b) repeated here along with a simple tree diagram:

(1.29) $Carlotta_{11}$'s dog accompanies $her_{11}/{*}herself_{11}$ to kindergarten.

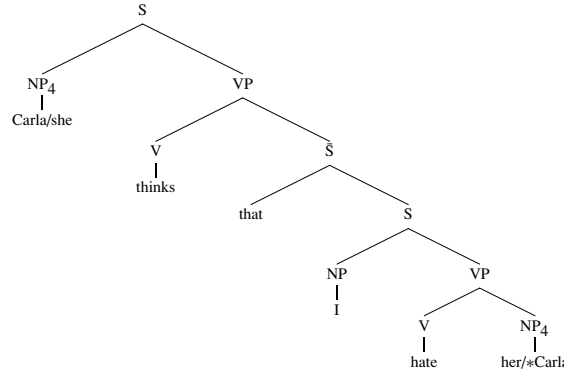
[3] Definitions in the literature usually include the qualification 'branching', even though, as Barker and Pullum [1990] and Pullum [1986] note, this is rarely argued for, nor required, by the data in any obvious way. The cases discussed in this book provide no exceptions to that; indeed the notion of semantic binding to be introduced in chapter 4 directly embodies Pullum's stricter and arguably more natural notion of IDV-command, according to which a constituent's c-command domain simply consists of its sister constituent(s).



Here, NP_{11}, *Carlotta*, does not c-command VP or anything therein: nodes dominating NP_{11} are NP_8 and S, which means that NP_{11} merely c-commands the $\bar{N}$ *dog*; VP, and the pronominal NP_{11} within it, though dominated by S, are not dominated by NP_8, which means they are not dominated by *every* branching node dominating NP_{11}, *Carlotta*, as is required for binding due to (1.26b). Accordingly, *[$_{NP}$ her(self)]$_{11}$* is not bound by *[$_{NP}$ Carlotta]$_{11}$* by the new definition of binding, especially (1.27c), so that the Binding Conditions correctly predict a non-reflexive (or a name) in that position.

Crucially, the new definition of binding is 'downward unlimited', because an NP that c-commands a node A also c-commands every node dominated by A. This is the key to handling the Binding Condition C cases. Consider again (1.22b), repeated here:

(1.30) ∗Carla$_4$/she$_4$ thinks that I hate Carla$_4$.



Similar to (1.21a), the matrix subject NP_4, *Carla*, c-commands the matrix VP, *and everything dominated by the matrix VP*, including the object NP_4. Since the subject NP_4 is also coindexed with the object NP_4 and precedes it, it qualifies as a binder. Binding Condition C then excludes a name as the object NP_4, while Binding Condition B allows a non-reflexive pronoun in that position.

This completes our introduction to the ABC of Binding Theory for English. It should be stressed that the Binding Conditions as stated above are no longer

about the traditional, intuitive concept of antecedence, but about a more abstract concept, binding. Binding Theory, so construed, is then a theory only about a subset of anaphoric relations, excluding non-c-command anaphora, both across and within sentences. This embodies a strong and non-obvious hypothesis, namely that c-command, or some other command notion (more about which is discussed in section 1.3), is of utmost significance for BT, and that, accordingly, the data fall into two broad natural classes – binding versus non-c-command anaphora. We will continue to reflect upon the validity of these hypotheses in the course of this book.

On the other hand, if Binding Conditions are indeed based on the notion of c-command, they can serve as a probe into the phrase structure of a sentence: if an NP blocks the occurrence of a coindexed pronoun or full NP', NP must c-command NP'. Binding Condition C in particular will be useful in this regard, since it applies across clause boundaries. It has been suggested, for example, that the pairs in (1.31) and (1.32) show that object clauses, but not temporal adverbial clauses, are c-commanded by the object, while both are c-commanded by the subject:

(1.31) (a) ∗The dog told him$_1$ [that the horse$_1$ would fall].
(b) The dog hit him$_1$ [while the horse$_1$ ate lunch].

(1.32) (a) ∗She$_8$'ll talk to me [when Sheila$_8$ gets back from lunch].
(b) I'll talk to Sheila$_8$ [when she$_8$ gets back from lunch].

While this method can be useful, it should be applied with care, for at least two reasons: first, as pointed out in section 1.2.3 above, the unacceptability of bound full NPs is itself not uncontroversial, and judgments seem to vary between speakers, but also in response to prosodic, stylistic, and discourse-pragmatic factors (see e.g. Carden and Dieterich [1981]; and Gerken and Bever [1986] for experimental results). Second, subordinated clauses are often found in displaced positions (e.g. through topicalization or extraposition), or at least could be for all we know, so that our conclusions from such examples rely in turn on our conclusions about the interaction of Binding Conditions with displacement (more on which in chapter 12). We will suggest that the phenomenon of *semantic binding*, to be introduced in chapter 4, may provide a more reliable diagnostic for c-command. Since we are presently concerned with demarcating the conditions on binding themselves (rather than presupposing them to figure out constituency), we will for the most part ignore constructions whose constituent structure is itself subject to debate.

1.2.5 Taking stock

It will be useful to separate several parts or components of the theory, as these will be subject to criticism, revision, or modification later, independent of each other:

- The **classification** of NPs according to their coreference and binding options. Here: three classes, reflexives and reciprocals ('anaphors'), non-reflexive pronouns ('pronominals'), non-pronominal or full NPs ('r-expressions').
- The identification of one or more **domain(s)** within which binding requirements apply. Here: the minimal clause.
- The formulation of a proper notion of **command** or *accessibility* as prerequisite for, and source of, asymmetry in binding. Here: precedence and c-command.

The general format of a Binding Condition can then be schematized as in (1.33):

(1.33) An NP of **class** must **(not)** be coindexed with a **commanding** NP within its **domain**.

Developing a general theory of binding is to formulate conditions of the general form in (1.33) for several languages, and in the process, to determine which exact values for the variables **class**, **command**, and **domain** are empirically most adequate, within a language, and cross-linguistically, and which are systematically irrelevant. In this book, we will be concerned with the notion of domain in chapter 3, the issue of classification in chapters 3 and 11, and the concept of command in section 1.3 of this chapter, as well as in chapter 12. Hopefully, the tripartition of the ingredients to the BT will prove useful in keeping track of the discussion.[4]

We have set up our system in such a way that each class of NPs may have (a) positive and/or negative Binding Condition(s) associated with it. Complementary distribution between two classes results if the negative binding domain for one element happens to be the same as the positive binding domain for the other. There is nothing in the formal system that accounts for the intuition one might have that one class of NPs is used *because* of the unavailability of the other, and vice versa. Intuitions of this kind have motivated so-called *blocking* approaches to Binding Theory, in which one form is used if (and perhaps only if) the other is excluded; examples of such approaches include Dowty (1980); Farmer and Harnish (1987); Huang (2000); and Levinson (1987, 1991, 2000); in light of the fact that non-complementary distribution between different pronoun classes is common across languages, including English (see chapters 3 and 11), we will not review these approaches further in this book; but see Burzio (1996, 1998); Kiparsky (2002) for refined, hybrid blocking approaches.

Exercise 1.2

A naïve approach to binding would be that the use of a pronoun is necessary to avoid repetition of full NPs, especially within a single sentence.

[4] Most authors appear to assume, implicitly, that the notion of command, once defined properly, is invariant across all Binding Conditions across all languages (but see the remarks at the end of section 1.3); if so, the variable **command** shouldn't be treated on a par with the others; we will leave this issue open in this book.

Use the distinction between antecedence and binding established above to make an argument against this view. Construct and provide crucial examples.

Exercise 1.3

Both sentences in (1.34) below could be used to illustrate that reflexives in English cannot be bound across a clause boundary. Explain how each does it. Can you think of reasons to prefer one mode of presentation over the other?

(1.34) (a) John_1 thinks that Bill_2 likes $\text{himself}_{2/*1}$.
(b) $*$John thinks that I like himself.

Exercise 1.4

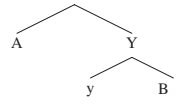
Consider the English possessive form *his*/*her*/*its own* (pretend that it is a single form). What requirements, if any, does it impose on its binder or antecedent? Formulate an appropriate Binding Condition for it, and adduce (acceptable and unacceptable) examples to support your proposal.

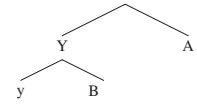
1.3 Command and precedence

As discussed in 1.2.5 above, the notion of *command* is one key ingredient in the formulation of Binding Conditions. We have used c-command plus precedence in our definition of binding above, hence as our relevant notion of **command** in (1.33). In this section we will critically reexamine these two notions, suggesting that they should perhaps be replaced with an altogether different command notion.

1.3.1 Against precedence

In our final definition of binding in (1.19) we have used two relations that give us an asymmetrical ordering among the NPs in a sentence, c-command and precedence. Is this necessary, or even tolerable? Consider the abstract phrase markers in (1.35):

(1.35) (a) 

(b) 

In (1.35a), the standard kind of case we have been looking at in English, A could bind B because A c-commands and precedes B; and B could not bind A, because it neither c-commands nor precedes A. In other words, according to our present definition of binding, (1.27), there are two different reasons why A can bind B, but not the other way around; the account is thus somewhat redundant, though perhaps harmlessly so.

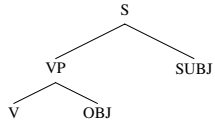
In (1.35b), on the other hand, A c-commands B asymmetrically, but B precedes A. If both factors are relevant, binding between A and B should be *ineffable*, i.e. impossible either way around, and regardless of the morphological shape of either NP. While English lacks clear instantiations of this configuration, other languages arguably provide them. Reinhart (1983a: 47) provides the following data from Malagasy (attributed to E. Keenan, p.c.), an Austronesian language spoken in Madagascar (for convenience, I have set the pronouns in boldface and the antecedents in italics):

(1.36) (a) namono **azy** ny anadahin- d- *Rakoto* (Malagasy)
hit/killed him the sister- of- Rakoto
'Rakoto's sister killed him.'
(b) ∗namono ny anadahin- d- *Rakoto* **izy**
hit/killed the sister- of- Rakoto he
'He killed Rakoto's sister.'

Malagasy is a VOS language, but, as the translations make clear, behaves rather like English with respect to the BT. In particular, (1.36b) appears to be a Binding Condition C violation, with the subject pronoun *izy*, 'he', illicitly binding the full NP *Rakoto* within the object NP. No such effect is found in (1.36a), in which the full NP is the subject, and the pronoun is in the object.

The data then suggest that the subject can bind the object, but not vice versa. Given that the object precedes the subject, this pattern straightforwardly prohibits a treatment in terms of precedence. C-command alone, however, would seem to provide the correct asymmetry, provided we assume that the basic clause structure of Malagasy is essentially as in (1.37) (cf. [1.35b] above):

(1.37)



We thus conclude that the inclusion of precedence in the definition of binding, while perhaps merely redundant for English, is actually harmful if we want to apply the notion cross-linguistically, and should be dropped.[5]

1.3.2 Limitations of C-command

Unfortunately, c-command is not unproblematic either. Languages that display flexible constituent ordering abound with examples in which a bound element precedes and c-commands its binder. Japanese, Korean, and German provide three examples. They are strictly or mostly left-branching languages with

[5] I found exactly one case in the literature in which precedence does seem to play a role, namely Samoan, as discussed in Chapin (1970); and Keenan and Stabler (1995). Whether this requires inclusion of precedence in the BT for Samoan, or can be captured in any other way, will be left as an open question here.

free word order, in all of which we find, for example, (reflexive-like) pronouns preceding coreferring full NPs:[6]

(1.38) **Zibun** -o *Hanako* -ga utagatte iru (Japanese)
self -ACC *Hanako* -NOM *doubts*
'Hanako doubts herself.'

(1.39) **Caki casin** -eke *Kim* -ŭn silmanghaŏssta (Korean)
self -DAT *Kim* -TOP *disappointed*
'Kim was disappointed in himself.'

(1.40) Oft hat **sich** *der Mann* im Spiegel betrachtet. (German)
often has self the man in the mirror watched
'Often the man watched himself in the mirror.'

These data are problematic for our definition of binding, because, unlike in the case of Malagasy discussed above, there is agreement that none of Japanese, Korean, or German has a constituent structure in which the reflexive objects in these examples are c-commanded by their subject antecedents.

It should be noted, however, that in all these examples, what could be called the 'logical subject' binds the 'logical object', regardless of order or c-command. In other words, one could conjecture that some more abstract level of representation displays an asymmetric ordering among constituents, at which all of the examples discussed in this section show the relative ordering in (1.41):

(1.41) 'subject' $\succ$ 'indirect object' $\succ$ 'direct object' $\succ$ 'prepositional object'

I have scare-quoted the notions 'subject', 'object', etc. because these are merely place-holders for whatever theoretical constructs a particular theory regards as relevant here. We will inspect several instantiations of such a hierarchy in turn.

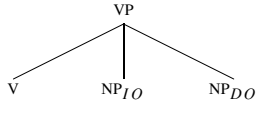
Before that, I want to bring up another set of data on which the c-command condition, even if it doesn't fail as spectacularly as on those in (1.38–1.40), seems deficient, namely so-called non-configurational or 'flat' structures. It is, for example, generally accepted that the objects in double object constructions in English show asymmetries in binding behavior:[7]

(1.42) (a) I showed John himself (in the mirror).
(b) $*$I showed himself John (in the mirror).

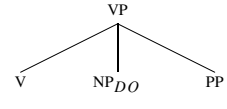
(1.43) (a) I showed Mary to herself.
(b) $*$I showed herself to Mary.

Suppose that the structure of the VP in double object constructions is as in (1.44):

(1.44) (a)



(b)



[6] Korean data from Keenan (1988):131.
[7] Reflexive examples from Barrs and Lasnik (1986):347 and Larson (1988):338.

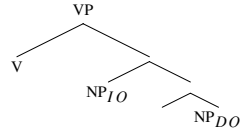
Given this structure, no asymmetries in binding are expected, contrary to fact, since the two objects mutually c-command each other, at least in the double NP case. Similar remarks apply, for example, to nouns with more than one post-nominal attribute:[8]

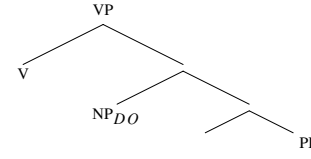
(1.45) (Ja čital) stat'ju *Tolstoja* o **sebe**. (Russian)
I read article-ACC *T.*-GEN *about himself*-LOC
'(I read) an article of Tolstoj about himself.'

Again, the immediately post-nominal phrase can bind the second one, but not vice versa. But, as in the case of verbal double-object constructions, the constituent structure appears to be flat.[9]

Both these examples would be treated correctly if we assumed c-command *and* precedence as prerequisites for binding, but, in the light of the problems with precedence encountered earlier, it is worth while to look for alternative treatments. A hierarchy along the lines of (1.41) above can provide such a treatment, as it provides a total ordering even among NPs that mutually c-command each other.

To be sure, there have been attempts to reconcile the data in (1.42–1.43) with a c-command approach to binding, notably Larson (1988), which argues that the asymmetries rather show that these structures are more complex than (1.44) and have the essential properties of those in (1.46):

(1.46) (a)


(b)


We do not have to elaborate on the details of this analysis (like the identity of the unlabelled nodes in [1.46]). The important thing is that a structure like (1.46) predicts the binding asymmetries as a function of asymmetrical c-command, without reference to anything else.

On the other hand, we saw the need to reconsider the c-command condition anyway in the light of the Japanese/Korean/German cases, and the two alternatives to c-command presented below are intended to capture those and 'flat structure' cases by the same mechanism, namely an independent, non-phrase-structural hierarchy.

This is obviously not the place to rule a final verdict on the (non-)existence of flat structures, but it bears mentioning that the binding facts alone can be treated by a refined command notion, which is presumably required independently, and that structures like in (1.46) are not necessitated by them. We will look at two such notions in what follows.

[8] Data based on Rappaport (1986):106.
[9] Note that the simplest binary branching structure would actually be *[[V/N XP_1] XP_2]*, which would yield the opposite asymmetries in c-command from what we find with the binding data.

1.3.3 Θ-command

Let us start with the perhaps most attractive version of a non-phrase-structural ordering among the NPs, namely the idea that this ordering is essentially semantic in nature, more precisely that it is based on the semantics of the *thematic relations* involved. An early and representative example of this approach is Jackendoff (1972):148, where it is proposed that the relevant asymmetric ordering among elements is derived from the hierarchy of *thematic roles* (or Θ-roles for short) in (1.47):

(1.47) Agent ≻ Location, Source, Goal ≻ Theme

The fact that *der Mann* can bind *sich* in the German example (1.40) above is then due to the fact that *der Mann* bears the Θ-role 'Agent,' which outranks the role of *sich*, 'Theme' on the Θ-hierarchy in (1.47). Likewise, *Rakoto* must not be coreferent with *izy* in the ungrammatical Malagasy example in (1.36), due to the fact that *izy* has the 'Agent' role in that sentence, which is higher than any other role in the Θ-hierarchy. The initial appearance that binding asymmetries correlate with a hierarchy among phrase-structure positions (as established by c-command) is merely an epiphenomenon of the dependence of phrase structure upon this same thematic hierarchy.

There are two more immediate advantages of Θ-command over c-command. First, as it is defined over thematic roles, it captures certain parallelisms between NPs and PPs. Not only can binding proceed 'into' PPs – a fact captured by c-command and Θ-command alike – sometimes it can also proceed 'out of' PPs, if the pertinent thematic command obtains:

(1.48) (a) We talked to John about himself.
(b) ∗We talked to himself about John.
(c) ∗We talked about John to himself.

These examples (from Wilkins [1988b]:208) show that Goal (as realized by a complement of *to*) Θ-commands Theme (as realized by a complement of *about*) and can therefore bind it, regardless of the fact that *John* doesn't c-command *himself*, because the first node dominating it is the PP.

Second, Θ-command allows a straightforward account of binding with so-called *psych-verbs*. With these verbs, a phrase-structurally lower argument appears to bind into a higher argument:[10]

(1.49) (a) Stories about herself generally please Mary.
(b) Each other's health worried the students.

While obviously unexpected from a phrase-structure point of view, these examples can be captured under the assumption that the role borne by the subject of

[10] Examples from Pesetsky (1987):127.